

Week 4

Date :- 27 july to 1 aug

Day :1

Objective

Understand the fundamentals of Node.js and set up a backend development environment.

Topics Covered

- **Introduction to Node.js**
- **Node.js Architecture**
- **Event-Driven Programming**
- **npm (Node Package Manager)**
- **Modules**
- **File System (fs)**
- **Package Management**

Practical Work

- **Installed Node.js and npm**
- **Initialized a new Node.js project using npm**
- **Created a simple HTTP server**
- **Explored built-in modules**

Day :-2

Objective

Learn how Express.js simplifies backend development.

Topics Covered

- **Introduction to Express.js**
- **Express Application Structure**
- **Routing**
- **Middleware**
- **Request & Response Objects**
- **Static Files**

Practical Work

Developed:

- **Home Route**
- **About Route**
- **Contact Route**
- **JSON Response API**

Day 3:-

Objective

Understand how REST APIs work and build API endpoints.

Topics Covered

- **REST Architecture**
- **HTTP Methods (GET, POST, PUT, DELETE)**
- **Route Parameters**
- **Query Parameters**
- **API Design Principles**
- **JSON Data Exchange**

Practical Work

Created a Student Management REST API with endpoints for:

- **Add Student**
- **View Students**
- **Update Student**
- **Delete Student**

Day 4:-

Objective

Learn database management using MongoDB.

Topics Covered

- **Introduction to MongoDB**
- **NoSQL Databases**
- **Collections & Documents**
- **CRUD Operations**
- **Mongoose**
- **Schema & Models**

Practical Work

- **Installed MongoDB**
- **Connected Express application with MongoDB**
- **Created Student Schema**
- **Inserted and retrieved records**

Day 5:-

Objective

Implement secure user authentication.

Topics Covered

- **Authentication vs Authorization**
- **User Registration**
- **Login System**
- **Password Hashing (bcrypt)**
- **JSON Web Token (JWT)**
- **Protected Routes**

Practical Work

Developed:

- **User Registration API**
- **Login API**
- **Password Encryption**
- **JWT-based Authentication**

Day 6,7:-

Objective

Test backend APIs and integrate them with the frontend.

Objective

Consolidate backend concepts by developing a complete full-stack module.